UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | PRE CALCULUS |
| COURSE CODE | MAT133 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 2 HOURS |

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of four (4) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 5 printed pages*

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## QUESTION 1

**a)** Solve the inequality $(5-x)^2 \leq 9$ and express the solution in real number line.
(5 marks)

**b)** Write the equation of circle $x^2 + y^2 - 4x - 2y - 11 = 0$ in standard form. Hence, sketch the circle with the correct center and radius.
(5 marks)

## QUESTION 2

**a)** Solve the inequality $\frac{2}{-2x-1} \leq 2$ . Express the answer in real number line.
(5 marks)

**b)** If $Z_1 = 2 - 7i$ and $Z_2 = -2 + 3i$ , state $\frac{\bar{Z}_1 i^{23}}{Z_2}$ in standard form $a + bi$ .
(5 marks)

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QUESTION 3

**a)** Given that $p(x) = \frac{7}{x} + 10$ and $q(x) = \sqrt{x-5}$ .

i) State the domain of $p(x)$ and $q(x)$ .
(2 marks)

ii) Determine $(q \circ p)(x)$ .
(3 marks)

**b)** Given that $f(x) = \frac{4-3x}{5x-7}$ , find the value of $x$ if $f^{-1}(x) = 2$ .
(5 marks)

QUESTION 4

**a)** Given $f(x) = \sqrt{x-9}$, $x \geq h$ and $g(x) = \frac{3-2x}{5+x}$, $x \neq k$. Find

i) the values of $h$ and $k$
(2 marks)

ii) $(f \circ g)(x)$
(3 marks)

iii) $g^{-1}(2)$
(4 marks)

**b)** Use long division to find the quotient and remainder when $P(x) = 2x^3 - x$ is divided by $(x-3)$.
(5 marks)

**c)** Solve the equation $5^{x+1} - 5^{2x} = 4$.
(6 marks)

**d)** Sketch the graph of $y = |x|$. Hence, sketch the graph of $y = 5 - |x-5|$ by using transformation techniques.
(5 marks)

END OF QUESTION PAPER

# APPENDIX 1

Formula sheet (as supplied with the paper).

1. $\sin^2 A + \cos^2 A = 1$
2. $\sin(A + B) = \sin A\cos B + \cos A\sin B$
3. $\cos(A + B) = \cos A\cos B - \sin A\sin B$
4. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$
5. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
6. $\sin 2A = 2\sin A\cos A$
7. $\tan^2 A + 1 = \sec^2 A$
8. $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$
9. Arc, $s = r\theta$
10. Area of sector $= \frac{1}{2}r^2\theta$
11. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
12. $a^2 = b^2 + c^2 - 2bc\cos A$
13. Area $= \frac{1}{2}ab\sin C$
14. Area $= \sqrt{s(s-a)(s-b)(s-c)}$

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